TRINETRA AI

Automated Photo Identification and Classification for Traffic Violations Using Computer Vision

Team TRINETRA AI

Gridlock Round 2 – Submission

June 21, 2026

Agenda

1. Problem
2. Architecture
3. Prototype Demo
4. Coverage & Roadmap
5. Conclusion

The Problem

- Traffic surveillance cameras generate **huge volumes** of images daily.
- Manual review for violations is **slow, costly, and inconsistent**.
- Human inspection cannot scale with growing camera networks.
- Need: an **automated, AI-driven pipeline** that can detect vehicles, classify violations, read plates, and generate enforceable evidence – with zero manual intervention.

Our Pitch

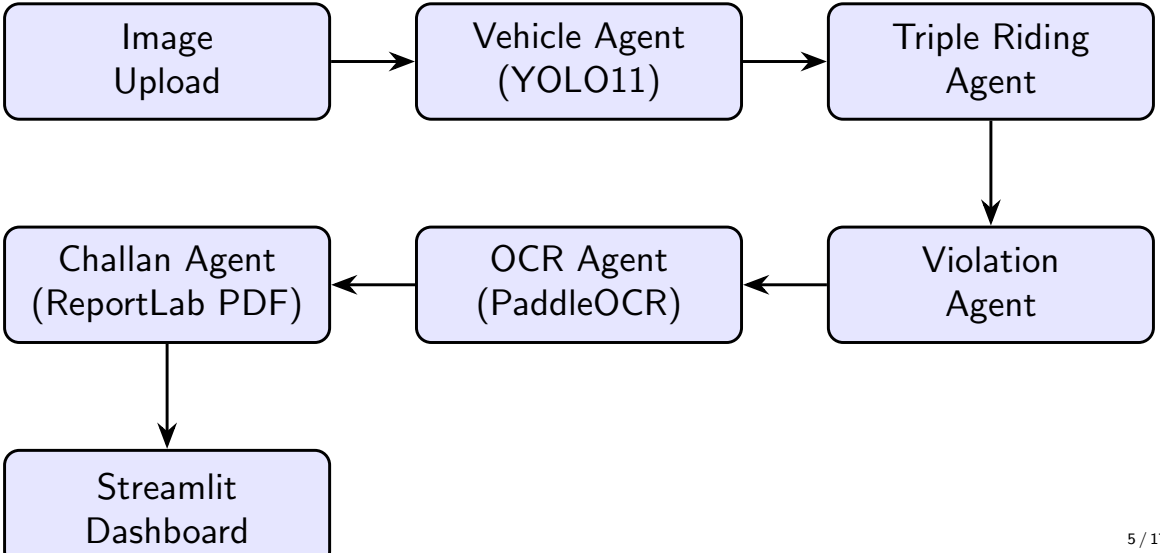
TRINETRA AI

A multi-agent computer vision pipeline that turns a single traffic scene image into a fully-formed, official PDF challan – automatically.

One-line idea

Detect → Identify Violation → Read Plate → Generate Evidence → Issue Challan – all in one pipeline, one image, no human in the loop.

System Architecture – Multi-Agent Pipeline



What Each Agent Does

Agent	Role
Vehicle Agent	YOLO11 detects persons, motorcycles, cars, buses, trucks
Triple Riding Agent	Flags violation if motorcycles ≥ 1 & persons ≥ 3
Violation Agent	Maps detected conditions $\rightarrow$ violation labels
OCR Agent	PaddleOCR reads license plate (full image, then per-vehicle crop)
Challan Agent	Builds PDF ticket with ReportLab: plate, time, evidence image

Stub agents (designed, planned for expansion): `helmet_agent`, `tracking_agent`, `enhancement_agent`, `plate_agent`

Tech Stack

Component	Library / Tool
Object Detection	Ultralytics YOLO11
License Plate OCR	PaddleOCR + PaddlePaddle
PDF Generation	ReportLab
Web Dashboard	Streamlit (dark mode)
Image Processing	OpenCV, Pillow
DL Backend	PyTorch, PaddlePaddle (CPU/GPU auto-switch)

Demo – Dashboard UI

[SCREENSHOT]

Streamlit dashboard: image upload, metric cards, violation badges

Demo – Detection & Annotated Evidence

[SCREENSHOT]

YOLO bounding boxes drawn on original image (Original vs Annotated tabs)

Violation Detection Logic

Current Prototype: Triple Riding

Condition: motorcycle_count ≥ 1 **AND** person_count ≥ 3 on the same motorcycle $\Rightarrow$ flagged as **Triple Riding**.

Architecture is rule-based and **pluggable** – new violation types slot in as new agents without touching the core pipeline:

- Helmet non-compliance
- Seatbelt non-compliance
- Wrong-side driving
- Stop-line / red-light violation
- Illegal parking

License Plate Recognition

Two-pass ANPR strategy:

1. Run PaddleOCR on the **full image**, regex-match Indian plate format `XX00XX0000` (e.g. `DL01AB1234`).
2. If no match $\rightarrow$ crop each detected vehicle's bounding box and re-run OCR **per crop** to boost accuracy.

CPU by default; auto-switches to GPU if CUDA-enabled PaddlePaddle is available.

Violation Risk Score

Condition	Points
Triple Riding detected	+40
Each additional violation	+20
Maximum score	100 (capped)

Risk score surfaces directly on the dashboard as a single metric authorities can use to triage and prioritize review.

Output – Automated PDF Challan

[SCREENSHOT]

Generated PDF challan: receipt no., plate, violations, embedded evidence image

Contains: Receipt Number, Vehicle Plate, Date & Time, Violations List, Embedded Evidence Image.

Mapping to Challenge Requirements

Challenge Task	**Status**
Vehicle / road user detection & classification	**Implemented**
Triple riding detection	**Implemented**
License plate detection & OCR	**Implemented**
Evidence generation (annotated image + metadata)	**Implemented**
PDF challan / reporting output	**Implemented**
Helmet non-compliance	Planned (stub agent ready)
Seatbelt non-compliance	Planned
Wrong-side driving	Planned
Stop-line / red-light violation	Planned (needs multi-frame tracking)
Illegal parking	Planned (needs multi-frame tracking)
Image enhancement (low light / rain / blur)	Planned (stub agent ready)

Scalability & Future Roadmap

- **Agent isolation** keeps compute efficient – OCR Agent only fires when Vehicle Agent confirms a vehicle exists.
- Deployable on **edge devices** or scaled as **cloud microservices** for thousands of camera feeds.
- Next agents to activate: `helmet_agent`, `tracking_agent` (multi-frame, enables red-light/parking/wrong-side), `enhancement_agent` (super-resolution for low light/rain/blur), `plate_agent` (dedicated plate-region detector for higher OCR accuracy).
- Evaluation plan: **mAP** for detection, **Precision / Recall / F1** for violation classification, latency & throughput for scalability.

Expected Impact

- Removes manual bottleneck in reviewing surveillance footage.
- Standardizes ticketing – consistent rules, no human bias.
- Produces a ready-to-issue PDF challan with embedded proof, end to end.
- Modular design means new violation types are additive, not a rewrite.

Thank You

Questions?

github.com/Almaas-Chaudhary/gridlock_round_2